

Active Inference ~ Parr, Pezzulo, Friston ~ Chapter 3 ~ BookStream #002.0

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Moving on to chapter 3, the high road to active inference.

And so, recall that the high road is like the why.

So, Ali, please begin, just go as long as you want, on through the high road to active inference.

Okay, so the high road to active inference begin with the question

about the conditions for the persistence of, quote-unquote, things.

I mean, what would we expect for a thing to behave if it persists through time?

And that's why sometimes FAP is referred to as the theory of every space thing.

And it's not absolutely everything, but thing just in this sense of, I mean, persisting through time.

And the notion that allows for this thing to act as it persists through time is markup blanket.

So, markup blanket is one of the most essential ingredients of the high road approach to active inference.

And that's why this section opens up with describing how markup blanket allows to describe the situation of interest

through this conceptual and mathematical tool.

But one thing that I always point out in almost all our textbook cohorts is,

in this section, it doesn't, I mean, exactly describe markup blankets rigorously enough

in terms of its carving up the, I mean, state space and so on.

So, one thing I think is, can be clarifying in following all the discussions around markup blanket is to keep in mind that it's just a boundary in the state space.

So, it's not necessarily spatiotemporal boundary.

Although, in many cases, it manifests itself as spatiotemporal boundary,

such as the, I don't know, cell membranes or organelle membranes and so on.

But it's not necessarily the case.

So, I think it can be helpful to keep that in mind when we, everywhere we see markup blanket.

But, in a very, very simply, markup blanket is what allows for the agent or the system

to be statistically separated from its environment, as is shown here in figure 3.1.

So, basically, markup blanket, mathematically, is just the, I mean, the sensory and active states together.

It consists of both of those states.

And, it is statistically separates what happens externally or internally from each other.

So, in other words, internal states cannot observe or infer anything about the external states directly, but only through the markup blanket.

And, that's why it's essential to describe the systems in this way,

because, obviously, in any sparse coupled systems,

there isn't a possibility to observe the external states directly,

but only through those sensory and active states.

And, one other thing that I also believe can be a bit confusing in this picture is that,

yes, typo in the active states and sensory states,
because it's important to observe that active state only,
or the flow in the active states, more precisely,
only depends on internal and markup states or blanket states.

And, on the other hand, sensory states only depend on the external and blanket states.

So, μ and x in those two equations should be exchanged.

Yes.

Yeah.

Let me just unpack this.

Anything to add before we continue?

Yes.

So, this is what's known as the particular partition.

Partition.

And, that terminology was brought to the fore by Carl Fristen's famous monograph in 2019.

And, it's a little bit of a pun.

It's a particular partition, because this is just one way to partition agent from environment,
figure from ground.

And, we call what it partitions out, the blanket and internal states, as the particle.

So, we can think about the most inert, least cognitive particle is like a speck of dust doing Brownian diffusion.

But, also, there are more sophisticated kinds of cognitive particles that can include world models,
counterfactuals, what would happen if I did this,

all of those kinds of sophisticated cognitive operations that we can explore in active inference.

So, note the small typo that Ali mentioned.

Internal states, and also these edges, reflect causal possibilities amongst internal, external, and blanket states.

Blanket constituting the active U and sensory Y states.

So, these are map, not territory.

This is not a spatiotemporal articulation.

These are like causal maps of the world that may have to do with spatiotemporal boundaries, but are not simply that.

And, sensory data or sensory states flow onto internal states.

Internal states are involved in action selection, resulting in active states,
which have some causal consequence in the world, the generative process, the external states,
which results in different sensory states coming in again.

And so, the Markov blanket is what makes the internal and the external states conditionally independent,
and that can be thought of as just a no telekinesis, no telepathy clause.

The only way that information comes across internal and external states,
which are symmetrical with each other,

is through the boundary or the holographic screen or the Markov blanket.

You can continue on, Ali.

Okay.

So, the next section is about surprise minimization and self-evidencing.

So, again, this term self-evidencing is a common term in active inference literature, first proposed by Jacob Howie.

So, basically, it refers to how a system or agent, or, in other words, a particular state, can gather the evidence for its self-persistence or self-existence through time.

In other words, by inferring the state of the environment and comparing that inference with its internal states, or, in other words, with its generative model,

it's basically a way, I mean, an interpretation of that kind of inference is,

on the one hand, there is this dynamical interaction between the internal and external state, but we can somehow interpret that physical dynamics as engaging in the active inference, or as a model for the persistence of the agent through time.

So, that's basically what it refers to as self-evidencing.

And then we go through the subsection 3.3.1,

which sets the stage for the recent formulation of active inference as in mechanics,

which is to somehow, to interpret the active surprise minimization

as minimizing the action through the Hamiltonian principle of least action,

which is a variational principle.

And by variational principle is, what is meant here,

is just a computational or mathematical tool

that allows for the computation or the derivation,

I mean, specific derivations to happen.

So, it's not identical to scientific theories or scientific, I don't know, facts or observations.

It's just a tool, a principle, a mathematical tool.

So, again, one of the main misconceptions about free energy principle

is that it is unfalsifiable.

Obviously, if we see it in this way,

it doesn't make sense to say that a mathematical tool or principle

is unfalsifiable,

because it doesn't say anything about the empirical evidence

of the phenomena we're talking about.

So, that's why here it's important to understand

how the surprise minimization can be seen

as this kind of variational principle.

And then,

equation 3.1

draws the parallel between
the surprisal as defined in section,
in chapter 2, sorry,
and this chapter.
So, it kind of ties up
all the arguments provided
in the previous chapter with this one
and how everything comes together
in a single
formulation of active inference
that they're just the
two distinct approaches
to arrive at same destination.
And then,
we go
also,
sorry,
the other important
equation
and, again,
another key
equation here
is equation 3.2
which is
a kind of
sorry
yes, equation
I meant to say equation 3.2
not equation 3.1
sorry,
my bad.
So, yes,
the other section
is about
the relations
between inference,
cognition,
and stochastic dynamics

as
I mean,
all the previous discussions
around
how
the active inference
can be seen
as a kind of self-evidencing
through the
variational
principle
such as
FEP
comes together
to
reframe
the
previous discussions
we saw
in chapter 2
about
perception as inference
and action as inference
and
to see
the concepts
of variational free energy
and expected free energy
through the lens
of variational free energy
variational
principle
of least action.
so
it
unifies nicely
all the
material

from
chapters 2
and 3
into
something
that
I mean
not
necessarily
distinct
from each other
but
just two sides
of
awesome
yeah
I find
table 3.1
to be
very exciting
it
draws
together
statistical
physics
Bayesian
information
information
theory
and
cognitive
interpretations
so it's
kind of like
learn one
thing
learn many
things

statistical
physics
has been
talking about
minimization
of variational
free energy
for a long
time
and such
methods
are
absolutely
everyday
and
professionalized
in Bayesian
statistics
active inference
is using
exactly
just
that
to
describe
perception
and
action
and
so
on
so
that's
very
exciting
box
3.2
describes

free energy
in statistical
physics
and active
inference
I sometimes
joke that
we have
a few
kinds
of free
energy
we have
Tesla
like
electrical
power
should be
available
to everybody
for no
cost
we're not
talking about
that right
now
there's
Gibbs
free
energy
which is
the quantity
that makes
chemicals
or
thermochemical
reactions
irreversible

like
the
hydrolysis
of
ATP
and
when we're
talking about
variational
free energy
we're talking
about an
information
geometric
space
with
similar
dynamics
similar
kinetics
and
thermodynamics
but rather
than describing
the reaction
coordinates
of a
chemical
reaction
we're
thinking
about it
in terms
of
Bayesian
updating
and this
is all

called
the
Bayesian
mechanics
3.41
continues
on
with
variational
free
energy
3.42
goes
into
expected
free
energy
so we
see
this
a lot
F
variational
free
energy
that's
the
real-time
unfolding
sensory
flow
and then
G
expected
free
energy
and the
policy

planning
as
inference
section
3.5
concludes
with
a novel
foundation
active
inference
to
understand
behavior
and
cognition
and
it
describes
a few
features
of
active
inference
including
its
distinguishing
features
from a few
other ways
that people
have looked
at behavior
and
cybernetics
section
3.6
goes into

a bit
more
detail
on
models
policies
and
trajectories
having to
do
with
agency
and
policy
selection
3.7
reconciliation
of
inactive
cybernetic
and
predictive
theories
under
active
inference
again
the exact
kind
of
thing
that
there's
a whole
literature
on
and it's
always

amazing
to hear
everybody's
perspective
on
in the
textbook
groups
3.8
active
inference
from
the
emergence
of
life
to
agency
and
3.9
summary
you want
to just
give
any other
thoughts
that you
have on
chapter
three
again
it's
one of
the
probably
most
fundamental
chapters

and
I mean
the topics
covered in
this chapter
is
essential
or absolutely
essential
to
understand
everything
active
inference
related
but
specifically
the
discussions
in
section
3.6
onward
again
I believe
is really
important
to understand
the
broader
context
of active
inference
and how
it
relates
to
all the

other
theories
but
specifically
section
3.8
provides
a nice
view
of how
we can
use
the same
modeling
the same
theoretical
tools
to model
both
non-living
dynamical
systems
or sparse
couple
dynamical
stochastic
systems
and also
to
the
systems
that
has
agency
or
sentience
so
it's a

much
broader
theoretical
framework
that doesn't
restrict
itself
to
only
particular
kinds
of
systems
or
agents
and we'll
see
much more
elaboration
on that
in
the
later
wonderful
all right
that concludes
our
overview
on chapter
three